Growth-mindset micro-lessons for first-year engineering students: a randomised controlled trial

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Abstract

We randomised first-year engineering students to three short growth-mindset lessons or to study-skills lessons. The intervention raised first-semester GPA by 0.11 grade points.

1. Methods

Design: individually randomised controlled trial at two universities. Participants: 612 first-year engineering students were enrolled; 574 had a first-semester GPA and were analysed. Intervention: three 20-minute online growth-mindset lessons in weeks 1-3. Comparator: three online lessons of matched length on study-skills facts (active control). Primary outcome: first-semester grade point average (GPA, 0-4 scale).

2. Results

First-semester GPA was 0.11 grade points higher in the growth-mindset group (mean difference 0.11, 95% CI 0.02 to 0.20).

3. Limitations

GPA was missing for 38 students, and the two universities may not represent other engineering programmes.

Fictional paper generated to test evidence-table extraction. Authors, journal, data and results are invented.